Project Proposal

Adaptive Reinforcement Learning for Lane Following and Obstacle Avoidance in a Procedurally Varied Urban Driving Environment

Group Members

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Summary of the Goals of the Work

The goal of this project is to develop and evaluate a reinforcement learning based autonomous driving agent capable of following a road lane and avoiding obstacles in a simulated urban environment using Webots. The work will focus on the Webots City demo, where a vehicle must navigate while maintaining lane alignment and reacting safely to static and, if time allows, simple dynamic obstacles.

Rather than training a single agent for one fixed scenario, this project will investigate how different design choices influence learning quality, robustness, and generalization. In particular, the work will compare different reward formulations and reinforcement learning algorithms, and it will study whether procedural generation of obstacle configurations improves the adaptability of the learned policy to unseen scenarios. The central idea is that a compact but informative observation space combined with carefully shaped rewards may produce a more stable and efficient learning process than simpler alternatives.

The main reinforcement learning algorithm used in the project will be Proximal Policy Optimization (PPO), due to its stability and suitability for continuous control tasks. A second algorithm, Soft Actor-Critic (SAC), will also be tested for comparison. If development time allows, a simple hand-crafted baseline controller may also be implemented for reference.

The quality of the solutions will be evaluated through experiments measuring success rate, collision rate, completion time, lane deviation, and learning efficiency. The project aims not only to obtain a functional autonomous vehicle controller, but also to produce a small scientific study on how reinforcement learning design decisions affect autonomous driving performance in simulation.

Proposed Approach to Solve the Problem

1. Description of the Setup

The project will be implemented in the **Webots simulator**, using the **City environment** as the base scenario. The simulated task consists of an autonomous vehicle that must drive along a road by following the lane while avoiding collisions with obstacles. The environment will be adapted to support repeated reinforcement learning episodes with randomized initial conditions and controlled scenario variation.

The vehicle will be placed on a road segment with a visible lane marking, namely the yellow line referenced in the project topic description. The world configuration will initially include straight road segments and smooth curves, allowing the development of a stable baseline before testing more difficult situations. Static obstacles such as barrels or boxes will be placed on or near the lane.

Their number and placement will be controlled parameters of the environment. In more advanced stages of the project, one simple dynamic obstacle may also be considered if the baseline system is working reliably.

The main environment parameters to be controlled are:

- the initial spawn position of the vehicle along the road;
- the initial orientation of the vehicle, including slight angular perturbations;
- the number of obstacles present in each episode;
- the position of obstacles within valid regions of the road;
- the maximum episode duration, measured in simulation steps;
- the difficulty of the scenario, determined by road geometry and obstacle density.

The project will begin with simpler scenarios, with static obstacles and a reduced level of environmental variation, in order to ensure a feasible and stable training process. More challenging scenarios will be introduced progressively.

The vehicle will use information derived from the simulator and from its virtual sensors. The proposed observation space for the main agent will be a compact, structured state representation including:

- lateral distance to the lane center or target lane line;
- heading angle error relative to the road direction;
- current vehicle speed;
- previous steering command;
- a set of frontal LiDAR distance readings;
- minimum distance to the nearest frontal obstacle;
- progress along the road segment.

Regarding the sensor configuration, the LiDAR will initially be used only through a frontal subset of rays, since frontal obstacle detection is the most relevant information for lane following and obstacle avoidance in this task. The number of rays effectively used by the agent, their angular coverage, and the normalization of the sensor readings will be defined empirically during the first implementation phase. If time allows, an extended observation space combining LiDAR based features with camera input may also be explored.

The action space will be **continuous**, allowing the agent to output:

- a steering command;
- a throttle or acceleration command.

These control signals will be normalized to fixed intervals before being mapped to the vehicle actuators. A braking component may be introduced if necessary, but the initial implementation will prioritize a simpler control formulation to reduce training instability. The actuator configuration will therefore be defined by the steering range, the acceleration command range, and the control update frequency used during simulation.

To integrate reinforcement learning with Webots, the simulator will be wrapped in a custom environment inspired by the Gymnasium interface. Each episode will consist of:

- resetting the vehicle to a valid randomized initial state;
- generating an obstacle configuration;
- simulating the agent's actions over time;
- updating the observation and reward at each control step;
- terminating the episode when the vehicle reaches the end of the track, collides, leaves the valid lane region, or exceeds the maximum number of time steps.

Training and testing will primarily be conducted on the group members' personal computers. If necessary, external computational resources such as Google Colab may be used for longer training runs and comparative experiments.

This setup is designed to provide a balance between realism, controllability, reproducibility, and manageable computational cost.

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2. Algorithms Planned and Why They Were Chosen

The main algorithm chosen for this project is **Proximal Policy Optimization (PPO)**. PPO is a strong candidate for this problem because it is known for stable training behavior, efficient policy updates, and good empirical performance in continuous control tasks. Since the autonomous vehicle must continuously adjust its steering and acceleration, PPO is well suited to learn smooth control policies.

In addition to PPO, the group plans to evaluate **Soft Actor-Critic (SAC)** as a second reinforcement learning approach. SAC is also designed for continuous action spaces and is often effective in problems where exploration and control smoothness are important. Comparing PPO and SAC will allow the group to assess whether the final performance depends significantly on the learning algorithm or more strongly on environment design choices, such as the reward function and observation space.

The project will not initially prioritize value based discrete action methods such as DQN, because the driving task is more naturally formulated with continuous controls. However, if time allows, a simplified discrete control baseline or a simple hand-crafted controller may be implemented for reference.

Beyond the algorithm itself, the project includes two key methodological ideas.

Reward shaping

A carefully designed reward function will be used to encourage desirable behavior. The initial dense reward function will combine:

- positive reward for forward progress along the road;
- positive reward for staying aligned with the lane direction;
- penalty for large lateral deviations from the lane;
- penalty for unsafe proximity to obstacles;
- penalty for abrupt steering changes;
- strong penalty for collisions or leaving the valid road area;
- terminal bonus for successfully completing the track.

A second, simpler reward configuration with sparser feedback will also be tested. This will allow the project to study whether dense reward shaping improves learning speed and final performance.

Procedural generation of obstacles

Instead of always training on a single fixed scenario, the group plans to implement procedural generation of obstacle placements within valid regions of the road environment. This is an important innovation in the project, because it makes the task more variable and allows the group to study robustness and generalization. An agent trained under more diverse obstacle configurations may perform better on previously unseen situations than one trained on a single repeated setup.

These algorithmic and design choices make the project more than a simple controller implementation. They turn it into a small research study on the relationship between environment design and reinforcement learning performance in autonomous driving.

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3. Proposed Experiments to Evaluate the Quality of the Approach

The experimental evaluation of the project will focus primarily on **three core comparisons**, which are considered essential to the main goals of the work. In addition, if time allows, **two complementary experiments** will be conducted to further analyze the learned policy.

Core Experiment 1: Dense Reward vs. Sparse Reward

This experiment will compare two reward formulations:

- a dense reward function with multiple behavioral components;
- a simpler sparse reward formulation with limited feedback.

Objective:

To evaluate whether reward shaping improves training stability, learning efficiency, and final navigation performance.

Performance metrics:

- success rate (% of episodes completed without collision);
- average completion time (s);
- average number of collisions per episode;
- average lateral lane deviation (m);
- mean episode reward.

Core Experiment 2: PPO vs. SAC

This experiment will compare the two main reinforcement learning algorithms under the same environment and observation settings.

Objective:

To determine which algorithm is more effective and stable for continuous lane following and obstacle avoidance in this simulator.

Performance metrics:

- success rate (%);
- average completion time (s);

- average number of collisions per episode;
- number of training episodes required to reach a predefined success threshold;
- variance of performance across different random seeds.

Core Experiment 3: Fixed Obstacles vs. Procedurally Generated Obstacles

In this experiment, one agent will be trained and tested with a fixed obstacle layout, while another will be trained with randomized obstacle placements.

Objective:

To evaluate whether procedural variation during training improves generalization to unseen scenarios.

Performance metrics:

- success rate in unseen test scenarios (%);
- collision rate in unseen test scenarios;
- average completion time (s);
- average lane deviation (m);
- recovery rate after perturbed initial positions (%).

Complementary Experiment 1: Ablation of State Features

This experiment will remove selected components from the observation space, such as heading error, speed, or some LiDAR rays.

Objective:

To identify which features contribute most to successful behavior and which are less important.

Performance metrics:

- success rate (%);
- average completion time (s);
- collision rate;
- learning speed measured in episodes until stable behavior.

Complementary Experiment 2: Robustness Under Perturbed Initial Conditions

The vehicle will start from different valid positions and slight orientation offsets relative to the road.

Objective:

To test the ability of the learned policy to recover from imperfect initial conditions rather than only succeeding when perfectly aligned.

Performance metrics:

- success rate (%);
- time to recover lane alignment (s);
- number of off-lane events per episode;
- average lane deviation during the first part of the episode (m).

4. Initial Reward Function Proposal

The initial dense reward function will be based on the following components:

- **Progress reward:** positive reward proportional to forward progress along the track.
- **Lane alignment reward:** positive reward when the vehicle heading is aligned with the road direction.
- **Lane centering penalty:** penalty proportional to lateral distance from the desired lane center.
- **Obstacle proximity penalty:** penalty when the vehicle is too close to frontal obstacles.
- **Control smoothness penalty:** penalty for abrupt steering changes or unnecessarily aggressive actions.
- **Collision penalty:** strong negative terminal reward.
- **Off-road or lane loss penalty:** strong negative terminal reward.
- **Episode success bonus:** positive terminal reward when the vehicle finishes the road segment safely.

The exact weights of these terms will be determined empirically through preliminary testing. The group expects to begin with a stronger emphasis on progress and safety, then adjust the reward balance to prevent undesired behaviors such as oscillations, stopping unnecessarily, or driving too slowly to avoid penalties.

This reward design is expected to encourage the emergence of behaviors that are not only goal-directed but also smooth and safe.

Objectives for Each Upcoming Week

Week 1: March 27 to April 2

- Final definition of the project scope and tasks.
- Study of the Webots City demo and the available vehicle models.
- Selection of the vehicle, sensors, and road configuration.
- Definition of the initial observation space, action space, and episode termination conditions.

Week 2: April 3 to April 9

- Creation of the custom Webots environment for reinforcement learning.
- Implementation of the reset and step functions.
- Extraction of lane-related and obstacle-related state variables.
- Initial manual tests of vehicle actuation and sensor readings.

Week 3: April 10 to April 16

- Implementation of the first dense reward function.
- Training of a PPO baseline agent in a simplified environment without obstacles.
- Debugging of simulation issues and correction of unstable behaviors.

Week 4: April 17 to April 23

- Addition of static obstacles to the environment.
- Introduction of randomized spawn positions and orientations.
- Preliminary tuning of reward coefficients and training parameters.
- Collection of the first quantitative performance results.

Week 5: April 24 to April 30

- Implementation and training of the SAC agent.
- Execution of comparative runs between PPO and SAC.
- Analysis of training curves and identification of failure modes.

Week 6: May 1 to May 7

- Development of procedural obstacle generation.
- Training and testing under multiple obstacle configurations.
- Preparation of the experiment on fixed versus randomized obstacles.

Week 7: May 8 to May 14

- Execution of the complementary analysis on selected state features, if feasible.
- Robustness testing with perturbed initial states, if feasible.
- Consolidation of metrics, logs, tables, and plots.
- Preparation of material for the intermediate checkpoint.

Week 8: May 15 to May 21

- Revision of results obtained up to the checkpoint.
- Refinement of the best-performing configuration.
- Organization of the codebase and reproducibility material.
- Planning of the structure of the scientific paper.
- Final analysis of results.
- Execution of any remaining complementary experiments, if needed.
- Writing of the scientific paper in IEEE format.
- Final code cleanup and preparation of the instructions.txt file.
- Preparation of videos and visual material for the oral presentation.

List of Objectives to Be Accomplished and Presented in the Intermediate Delivery

By the intermediate checkpoint, the group expects to have completed and demonstrated the following:

1. A functional Webots-based simulation environment for lane following and obstacle avoidance.

2. A custom reinforcement learning environment compatible with training workflows.
3. A working PPO baseline capable of following the road in simple scenarios.
4. A first version of the reward function implemented and tested.
5. Static obstacle insertion and preliminary obstacle avoidance behavior in simple scenarios.
6. Preliminary training curves and performance metrics.
7. A short video demonstration showing both successful and failed runs.
8. A brief explanation of the next experimental comparisons, namely PPO versus SAC and fixed versus procedurally generated obstacles.

The intermediate delivery will therefore demonstrate that the project has already moved beyond planning and into functional development, while still leaving room for meaningful experimental analysis in the final phase.